11-10-2026

prerequisite

→ python (basic)

Agenda:

Python - 1 → variable, datatype, operators, collection, string operators, control-flow (if else), loop, function

programming language → language (commands) → computer → run → complete a task

Python → Machine learning
computer vision
Generative AI
Agentic AI
web development
API server

Store a value : variable → a variable is a name that refers to an object in python

3 ↑ 2 → a ↑ b → 5
a ↑ 2 →

a b

q = 3 → memory → int-obj → address

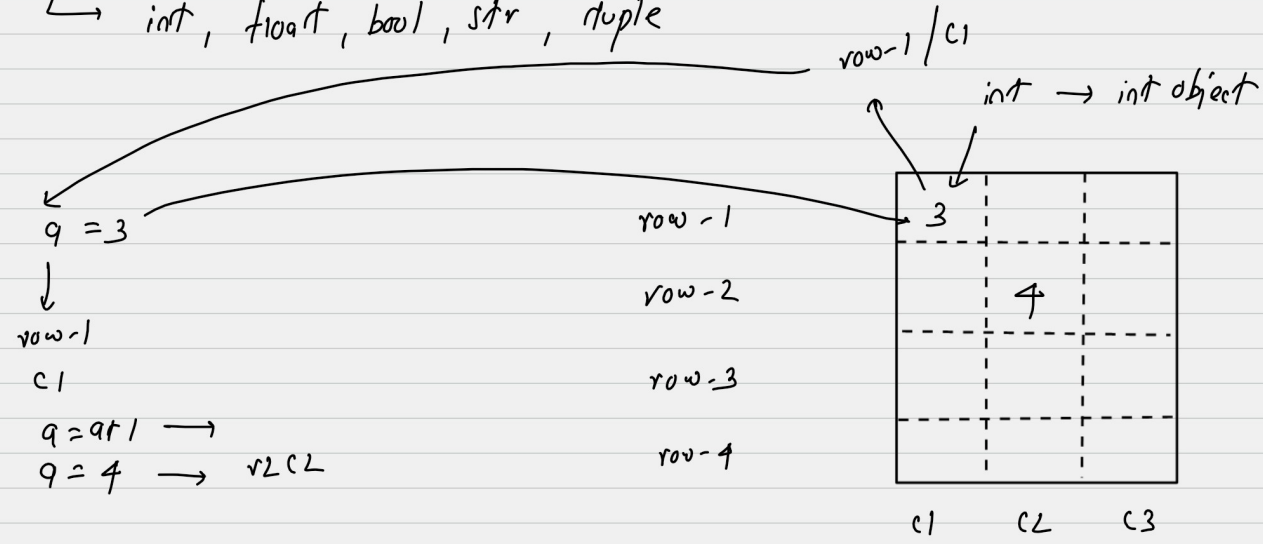
variable → name attached to a value

value
↓
types
- number → int, float, complex
- Boolean → True, False
- Text → string
- Sequence → list, tuple, range
- Set → set
- Mapping → dictionary
- Null Type → NoneType

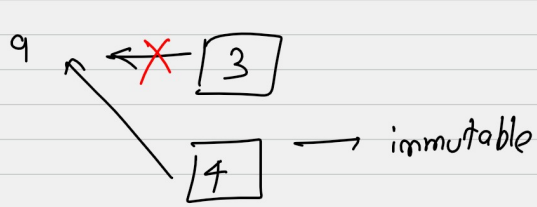
mutable vs immutable

mutable → objects/value can be changed after creation
↳ list, dict, set

immutable → objects cannot have their value changed after creation
↳ int, float, bool, str, tuple



3 → 4 → mutable



row-1 at c1
↓
number

---

id (variable)
↓
reference → numberic representation

---

primitive vs non primitive data type

Basic (primitive): int, float, bool, string NoneType

(non-primitive): list, tuple, set, dict

Operators

A operator is a symbol or keyword that perform an operation on value or variable.

addition operator

2 + 3

↑ val ↑ val

a = 2
b = 3
a + b → 5

Arithmetic operator

operators	Meaning	Example	Result
+	Adding	10 + 3	13
−	Subtraction	10 − 3	7
*	multiplication	10 * 3	30
/	True division	10 / 3	3.333
//	Floor division	10 // 3	3
%	Modulus (remainder)	10 % 3	1
**	Exponentiation	2 ** 3	8

Comparision operators

compare 2 values and return Boolean

operator	Meaning		
==	equal to	2 == 3 →	false
!=	Not equal to	2 != 3 →	True
>	Greater than	2 > 3 →	F
<		2 < 3 →	T
>=		2 >= 3 →	F
<=		10 <= 10 →	T

10 <= 10

<= → smaller → NO

<= → equal to → True / Yes

Assignment operator

Assign values to names or update existing value

→ `x = 10`

`x += 5`	`x = x+5`	`x = 15`
`x -= 2`	`x = x-2`	`x = 8`
`x *= 2`	`x = x*2`	`x = 20`
`x /= 2`	`x = x/2`	
`x //= 2`	`x = x//2`	
`x %= 3`	`x = x%3`	
`x**= 2`	`x = x**2`	

`x8=`

Logical operator → used to combine or reverse boolean conditions

And, Or, Not → True → 1, False → 0

AND

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT

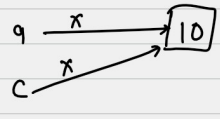
A	B
0	1
1	0

Identity operator

a = 10

c = 10

a is c → True



Bitwise operators (Binary)

12 → Bits → Binary

2^4 2^3 2^2 2^1 2^0 ← 4 bits → max → 15

5 bit → max → 31

16 8 4 2 1

8 + 4 → 12

1	1	0	0	– 12
0	1	0	0	– 4
0	0	1	1	– 3
0	0	0	0	– 0

Binary operator

	Meaning	Usage
&	AND	12 & 4
|	OR	12 | 4
~	NOT	
<<	Left shift	5 << 1
>>	Right shift	

1100 → 12

& 0100 → 4

0100 → number → 4

~ 1100 → 12

0011 → 3

0101 → 5

<<

01010 → 10

1111 → 15

<<1

11110 → 14

collection :

A collection store multiple values in a single object.

a = 3 → int

name_1 = " " 1 million ule
name_2 = " "

↓

Type	ordered	Mutable	Duplicate
list	Yes	Yes	Yes Allowed
tuple	Yes	NO	Yes
set	does not gurantee positional order	Yes	Not Allowed
dict	Yes	Yes	keys must be unique

list :

append (x) → add one item at the end
remove (x) → remove the first matching value
pop (index) → removes & returns an item
insert (index, value) →
index (value) → first matching index
count (value)
sort ( )

a = [1, 2, 3]
0 1 2

a.insert(1, 5)

a = [1, 5, 2, 3]
0 1 2 3

len (sequence)

index → position of value

sequence → str, collection

a = [1, 2, 3, 4, 5, 6] ← values

0 1 2 3 4 5 ← index

position starts with → 0 → left side

a → 0
→ 5

a[3] → 4

Tuple :—

A tuple is an ordered, immutable collection

a = (1, 2) → tuple

→ read
→ edit → ✗ → not allowed

→ (Day of week)
(M, T, W, T, F, S, S)
